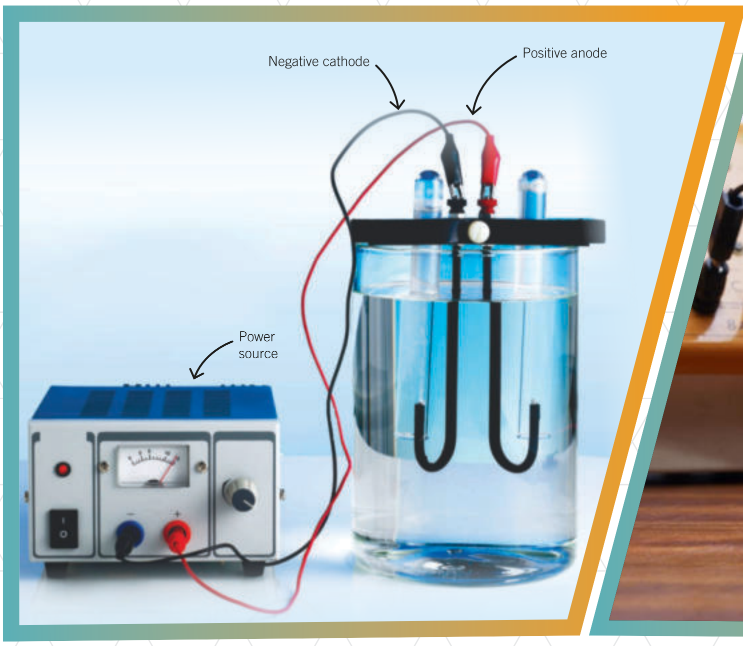
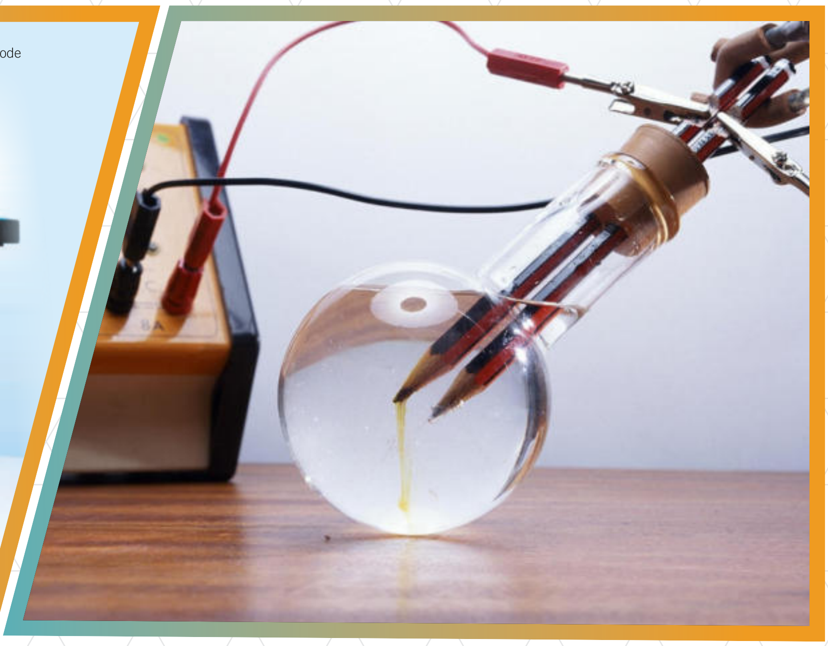
# UNIT 07

# ELECTROCHEMISTRY



**ELECTROCHEMISTRY:** is the study of the relationships between electricity and chemical reactions. It includes the study of both spontaneous and nonspontaneous processes.

## OXIDATION:

There are several definitions of oxidation & reduction:

➢ Oxidation is the gain of oxygen.

$$2Mg + O_2 \longrightarrow 2MgO$$

➢ Oxidation is loss of electrons by an atom or an ion.

$$H_2 \longrightarrow 2H^+ + 2e$$

➢ Oxidation is the increase in oxidation state.

$$Cu^0 \longrightarrow Cu^{+2} + 2e$$

➢ Oxidation is the loss of hydrogen.

$$H_2S + O_2 \longrightarrow SO_2 + H_2O$$


## REDUCTION:

➢ Reduction is the loss of oxygen.

$$CuO + H_2 \longrightarrow Cu + H_2O$$

➢ Reduction is gain of electrons by an atom or an ion.

$$2H^+ + 2e \longrightarrow H_2$$

➢ Reduction is the decrease in oxidation state.

$$Cu^{+2} + 2e \longrightarrow Cu^0$$

➢ Reduction is gain of hydrogen.

$$H_2S + O_2 \longrightarrow SO_2 + H_2O$$

## REDOX REACTION:

The reaction in which both oxidation and reduction reactions occur simultaneously.

$$H_2S + O_2 \longrightarrow SO_2 + H_2O$$

## OXIDIZING AGENT:

An oxidizing agent oxidises another substance – and is itself reduced or simply the chemical species which accepting the electrons. e.g. $H_2SO_4$, $HNO_3$, $KMnO_4$, $K_2Cr_2O_7$, $Cl_2$, $Br_2$, $I_2$ etc.

## REDUCING AGENT:

A reducing agent reduces another substance – and is itself oxidized or simply the chemical species which losing the electrons. e.g., Alkali metals, Al, $H_2S$, Zn, NaH, KH etc.

## ELECTROLYTE:

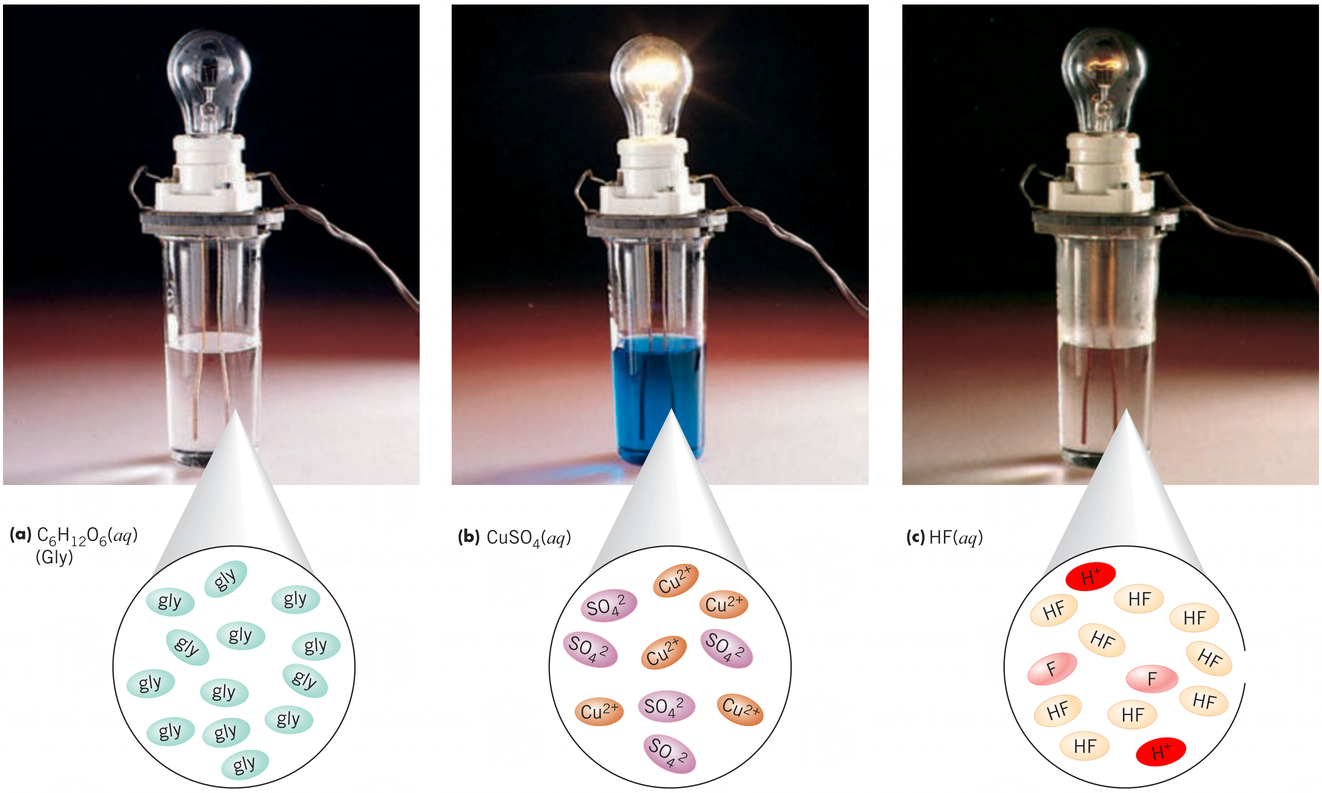
The substances, which can conduct electricity in their aqueous solutions or molten states, are called electrolytes. For example, solutions of salts, acids or bases are good electrolytes.

There are generally two types of electrolytes on the basis of their strength.

**(I) Strong Electrolytes:** The electrolytes which ionize almost completely in their aqueous solutions and produce more ions, are called strong electrolytes. Example of strong electrolytes are aqueous solutions of NaCl, NaOH and $H_2SO_4$, etc.

**(ii) Weak Electrolytes:** The electrolytes which ionize to a small extent when dissolved in water and could not produce more ions are called weak electrolytes. Acetic acid ($CH_3COOH$) and Ca $(OH)_2$ when dissolved in water, ionize to a small extent and are good examples of weak electrolytes.

**(iii) Non-Electrolyte:** The substances, which do not ionize in their aqueous solutions and do not allow the current to pass through their solutions, are called non-electrolytes. For example, sugar solution and benzene are non-electrolytes.




**Nonelectrolytes, Strong Electrolytes, and Weak Electrolytes** (a) A solution of glucose sugar ($C_6H_{12}O_6$) does not conduct electricity. (b) A $CuSO_4$ solution is a strong electrolyte and conducts electricity. (c) A hydrofluoric acid solution [HF(*aq*)] is a weak electrolyte and conducts a limited amount of electricity.

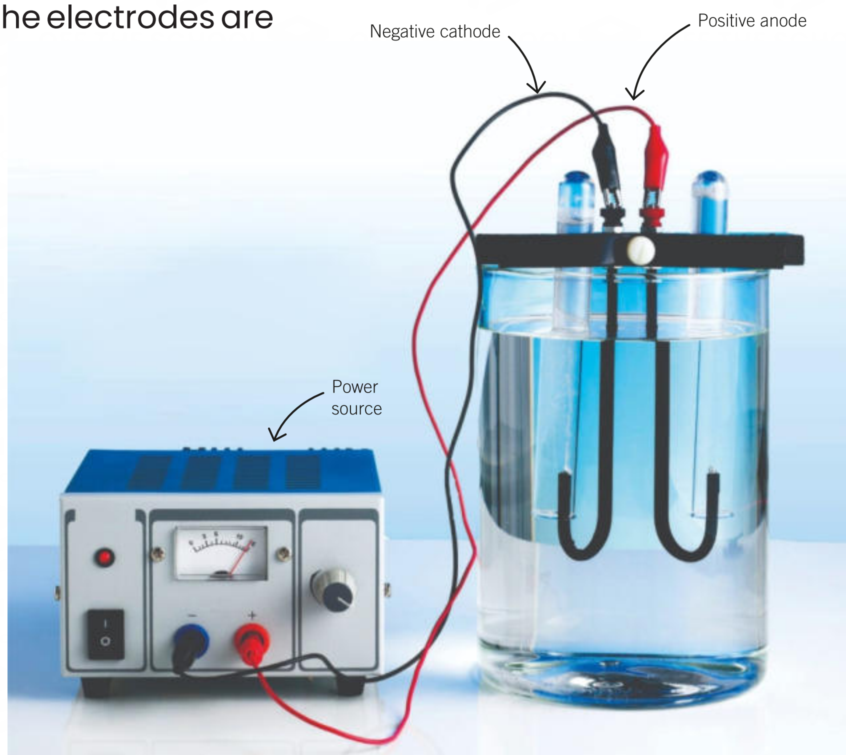
## ELECTRODE:

An electrode is a conductor used to establish electrical contact with a nonmetallic part of a circuit, such as an electrolyte. **OR** The electrodes are rods, made from either carbon (graphite) or metal, which conduct electricity to and from the electrolyte.

Generally, two types of electrodes are known;

**(i) Anode:** an electrode at which an oxidation half-reaction (loss of electrons) takes place.

**(ii) cathode:** an electrode at which a reduction half-reaction (gain of electrons) takes place.



## ELECTROCHEMICAL CELLLS:

The device which converts electrical energy into chemical energy and vice versa by using redox reaction.

**Explanation:**

An electrochemical cell consists of two electrodes. Each electrode is in contact with an electrolyte; the electrode and the electrolyte make up a half-cell. The two electrodes are connected by a wire, and a porous barrier or salt bridge separates the two electrolytes. Electrochemical cells are of two types:

**(i)** Electrolytic cells

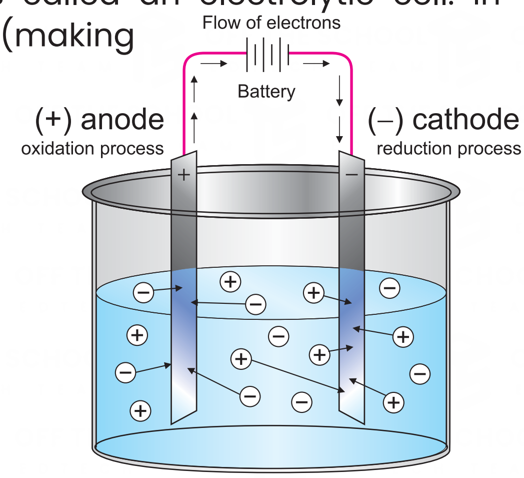
**(ii)** Galvanic or voltaic cell

## ELECTROLYTIC CELL:

A device in which an external source of electrical energy does work on a chemical system, turning reactant(s) into higher-energy product(s).

**Explanation:**

A reaction that is driven in this way, by the consumption of electrical energy, is called electrolysis and a cell in which electrolysis occurs is called an electrolytic cell. In electrolytic cells electrons are pumped into the cathodes (making them the negative electrodes), and out from the anodes (making them the positive electrodes).



Electrolytic cell

**Applications:**

- In preparation of sodium metal (down's cell).
- In preparation of $Cl_2$ gas (Nelsons cell).
- In extraction of Aluminum metal.
- In refining of copper.
- Electroplating..

## FARADAYS LAWS OF ELECTROLYSIS:

### Faraday's 1st law of electrolysis:

Intro: M.faraday was the bristish scientist and greatly contributed in the field of electrochemistry.

**Statement:**

The mass of a substance liberated at an electrode during electrolysis is directly proportional to the quantity of electricity transferred at that electrode.

$$W \propto A \times t$$

$$\mathbf{W = ZAt}$$

**W =** amount of deposited or liberated substance. (Grams or g)
**Z =** constant [Electrochemical equivalent (gram per coulomb or $g.C^{-1}$)].
**A =** current (Amperes or A).
**T =** time (Seconds or sec.)

**Electrochemical equivalent (Z)** is the weight of the substance collected at the electrodes when one coulomb of electric charge is passed through the electrolyte.

## Faraday's 2nd law of Electrolysis:

**Statement:**

The amount of different substances deposited or liberated due to passage of same quantity of current through different electrolytes are proportional to their chemical equivalent masses.

**Equivalent mass or weight:**

$$\text{Equivalent mass or weight} = \frac{\text{atomic weight}}{\text{valency}}$$

**Faraday:**

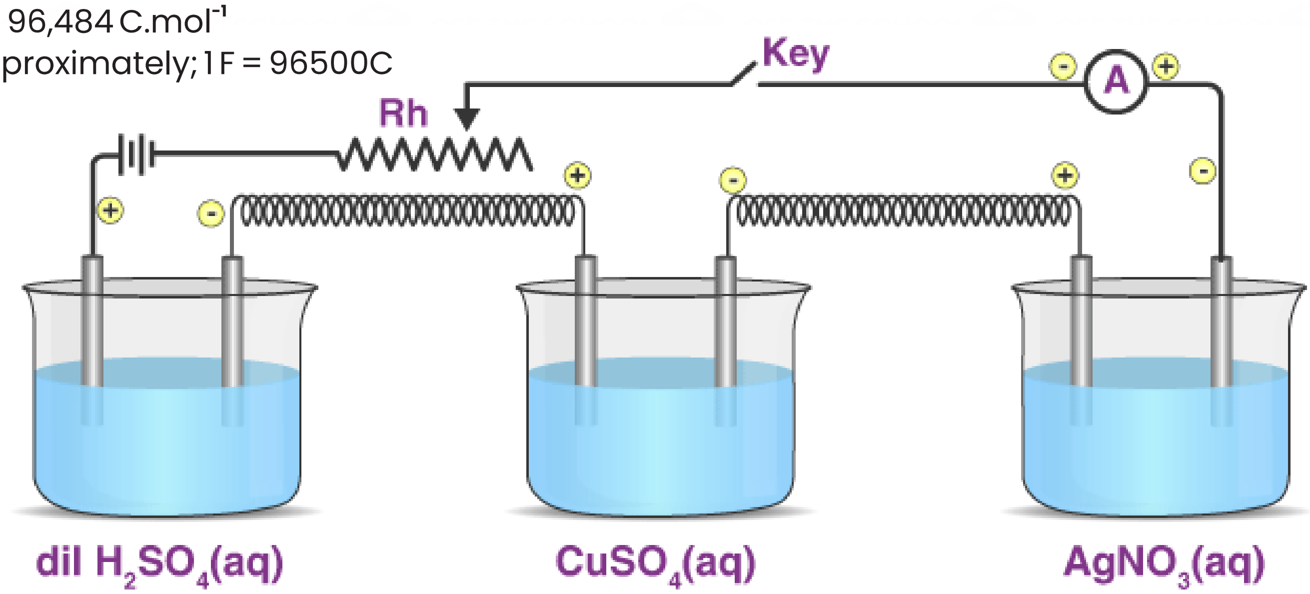
The basic unit of electric charge is faraday and it is defined as the charge on one mole of electron.

**F =** Avogadro's number x charge on electron in coulombs

**F =** $6.02 \times 10^{23}\ mol^{-1} \times 1.602192\ C$

**F =** $96{,}484\ C.mol^{-1}$

Approximately; 1 F = 96500C




## BATTERIES:

A battery is a device that stores energy and releases it as electrical power to make things needed to power your cell phone, watch, and calculator. Batteries also are needed to make cars start, and flashlights produce light. Within each of these batteries are voltaic cells connected in a series that produce electrical energy.

There are two types of batteries:

**1) Primary (Non-rechargeable) battery.**

Examples include:

## DRY-CELL BATTERIES:

Dry-Cell Batteries Dry-cell batteries are used in calculators, watches, flashlights, and battery-operated toys. The term dry cell describes a battery that uses a paste rather than an aqueous solution. Dry cells can be acidic or alkaline.

**Construction of Dry Cell**

- Outer shell: Zinc metal acts as the negative terminal and container.
- Electrolyte: Moist paste of ammonium chloride inside insulating material.
- Carbon rod: Runs through center, acts as positive terminal.
- Manganese dioxide: Surrounds carbon rod to prevent gas buildup (depolarizer).
- Separator: Keeps zinc shell and electrolyte apart to avoid short circuits.

**Working of Dry Cell**

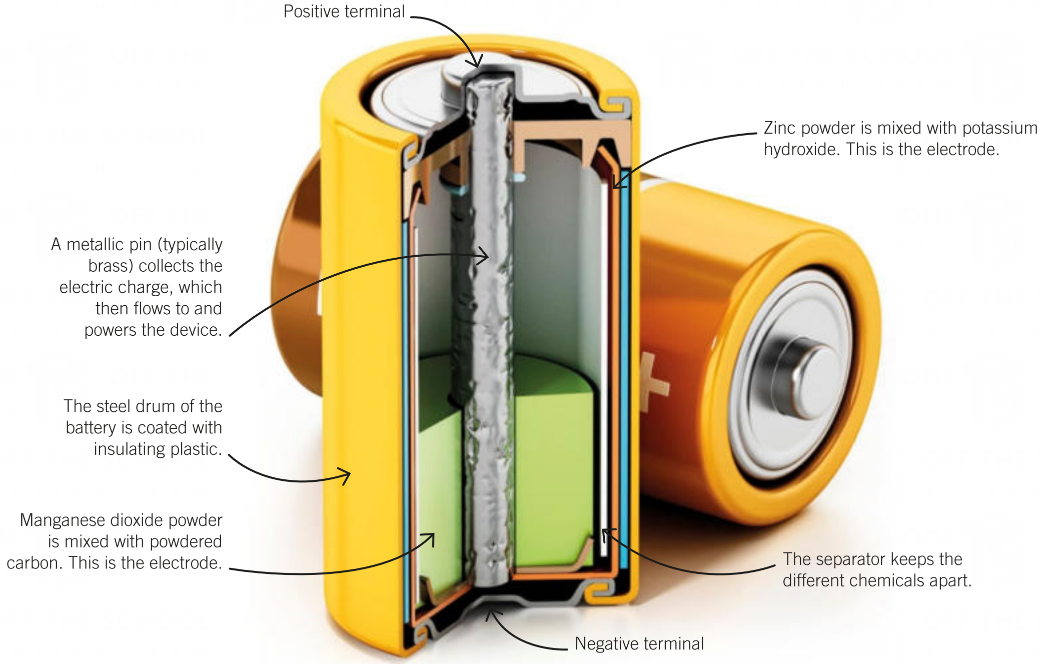
Zinc (negative terminal) loses electrons (oxidation):

$$Zn \rightarrow Zn^{2+} + 2e^-$$

Electrons flow through external circuit to carbon rod (positive terminal).

Manganese dioxide reacts with ammonium ions and electrons to stop hydrogen gas formation(reduction):

$$2MnO_2+2NH^+_4+2\ e^- \longrightarrow Mn_2O_3+2NH_3+H_2O$$

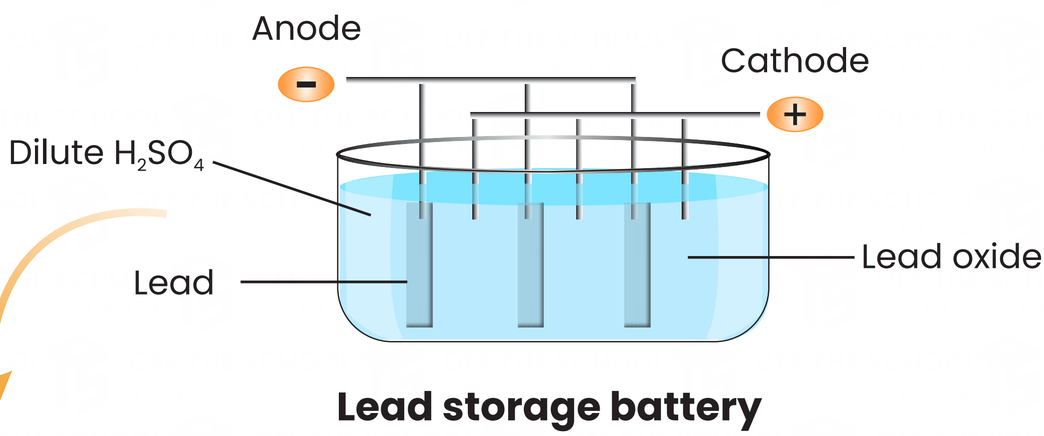



## LEAD STORAGE BATTERY (LEAD-ACID BATTERY):

A lead storage battery is a type of rechargeable battery. It has multiple cells connected together, with lead plates acting as electrodes. The anode is made of lead (Pb), and the cathode is made of lead oxide ($PbO_2$). These electrodes are placed in a solution of dilute sulfuric acid ($H_2SO_4$).

**Discharge:** During discharge, the lead plates react with sulfuric acid to form lead sulfate ($PbSO_4$) and water ($H_2O$).

**Charge:** During charging, lead sulfate ($PbSO_4$) is converted back to lead and lead oxide.This process can be reversed, allowing the battery to be recharged.




**Lead storage battery**

## ALLOY:

The homogenous mixture of metals is known as alloy.
For example, mixing molten zinc and copper gives the gold-colored alloy called brass. When this solidifies, it is hard, strong, and shiny. It is used for door locks, keys, knobs, and musical instruments such as trumpets. Turning a metal into an alloy changes its properties, and makes it more useful.

**Pure Gold vs. Gold Alloy** Pure gold is too soft to use in jewelry. Alloying it with other metals increases its strength and hardness, while retaining its appearance and corrosion resistance.


**(a)** 24-karat gold is pure gold.

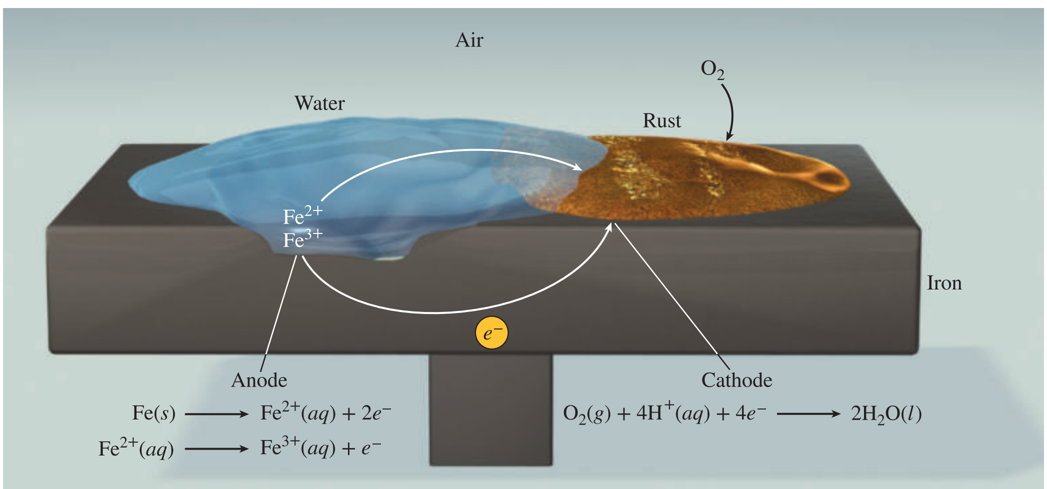
**(b)** 14-karat gold is an alloy of gold with silver and copper. 14-karat gold is 14/24, or 58.3%, gold.

## CORROSION AND ITS PREVENTION:

The term corrosion generally refers to the deterioration of a metal by an electrochemical process. There are many examples of corrosion, including rust on iron, tarnish on silver, and the green layer that forms on copper and brass.

### I) Rusting of iron:

Iron undergoes redox reaction in the presence of air or water to form hydrated iron (III) oxide ($Fe_2O_3.nH_2O$) called rusting of iron. Rusting requires oxygen and water.
The corrosion of iron and steel has a special name: rusting. The red-brown substance is formed that is rust.




The electrochemical process involved in rust formation. The $H^+$ ions are supplied by $H_2CO_3$, which forms when $CO_2$ from air dissolves in water.

### Prevention from corrosion:

- Alloying.
- Greasing.
- Galvanising. (Coating with zinc has a special name: galvanising.)
- Coating with paints.
- Cathodic protection.
- Metallic coating (electroplating).
- Tin Coating.

## ELECTROPLATING:

The process of deposition of metal at the surface of another metal is known as electroplating or metallic coating.

### Procedure for Electroplating:

In this process the object to be electroplated is cleaned with sand, washed with caustic soda solution and finally it is thoroughly washed with water. The anode is made of the metal, which is to be deposited like Sn, Zn, Ag, Cr, Ni. The cathode is made up of the object that is to be electroplated like some sheet made up of iron. The electrolyte in this system is a salt of the metal being deposited. The electrolytic tank is made of cement, glass or wood in which anode and cathode are suspended. These electrodes are connected with a battery. When the current is passed, the metal from anode dissolves in the solution and metallic ions migrate to the cathode and discharge or deposit on the cathode (object). As a result of this discharge, a thin layer of metal deposits on the object, which is then pulled out and cleaned.

### Electroplating of silver:

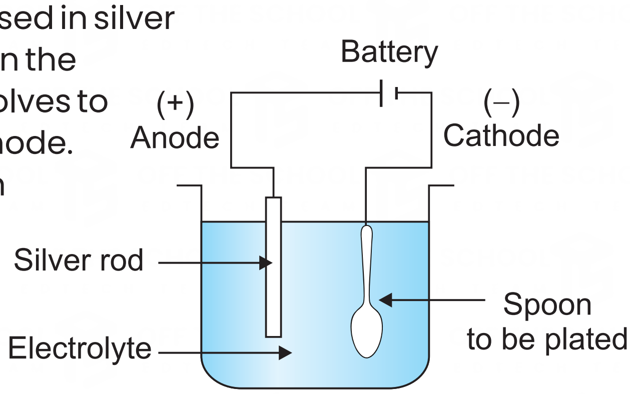
- Cathode ⟶ steel or any desired metal.
- Anode ⟶ sillver
- Electrolyte ⟶ aqueous solution of silver chloride ($AgCl_2$)

### Explanation:

The pure piece of silver strip acts as anode immersed in silver chloride solution. The cathode is steel spoon. When the current is passed through the cell, the anode dissolves to produce Ag+2 ions, that migrate towards the cathode. At cathode they are discharged and deposited on the spoon. The chemical reaction can be represented as:

**At anode:** $Ag \longrightarrow Ag^{+} + 1e$

**At cathode:** $Ag + 1e \longrightarrow Ag$




Electroplating of an object

### Tin plating:

- Cathode ⟶ steel spoon
- Anode ⟶ tin metal
- Electrolyte ⟶ acidified tin sulfate ($SnSO_4$)

### Zinc plating:

- Cathode ⟶ steel object.
- Anode ⟶ zinc metal
- Electrolyte ⟶ potassium zinc cyanide ($C_4K_2N_4Zn$)

### Chromium plating:

- Cathode ⟶ desired or base metal
- Anode ⟶ chromium metal
- Electrolyte ⟶ acidified chromium sulfate $Cr_2(SO_4)$